

VSEPR-SIM Research Platform

§L Five-Layer Vertical Integration Stack

Materials Discovery Engine
Full Fuel Cycle Simulation
Cradle to Product

Layer	Description
L1	Paper Identity — Formula, class, name. Static.
L2	Atomistic — Z , A , Q , ε . Physics and energy state.
L3	Molecular 3D — Geometry, bonding, coordination.
L4	Atomistic / CG — Beads, phase behaviour, bulk interactions.
L5	Macro — CAD-scale geometry, volumes, surfaces.

The particle identity from L2 propagates all the way up.
The iron in a pipe at L5 is still Fe with its Z , A , Q underneath.
The macro geometry is the accumulated consequence of everything below it.

Implementation files: `include/layer_stack.hpp`
`coarse_grain/models/layer_boundary.hpp`
Status: Research-oriented active development

Contents

§1 Architecture Overview

1.1 Purpose

The five-layer stack is the routing document for the entire VSEPR-SIM platform. It answers the question: *what is the particle at every scale, and how does it get there?*

The stack is not a layer model in the OSI sense — layers do not replace each other. They accumulate. Each higher layer is an interpretation of the same underlying physical object at a different resolution.

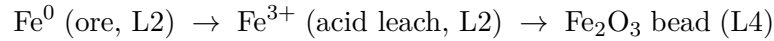
1.2 The Vertical Integration Invariant

Postulate 1.1 (Vertical Integration Invariant). For any particle i and any layer $k \in \{2, 3, 4, 5\}$:

$$L_k(i).Z = L_2(i).Z, \quad L_k(i).A = L_2(i).A$$

Nuclear identity is immutable across all layers under all non-nuclear transformations. The iron in a pipe at L5 is still Fe at L2.

Charge Q_i and energy state ε_i are *environment-dependent* and evolve as the chemical context changes:



Same $Z = 26$. Different Q , different ε , different Σ_i (structural role changes from metallic to ionic at oxidation).

1.3 The Compute Problem

The discovery-end use case requires pushing millions of candidates through the stack. Most die at L1. The rest are screened progressively:

Layer	Typical survivors	Cost per candidate	Gate criterion
L1 screen	10^6	μs	Formula valid, class not excluded
L2 screen	10^4	ms	ε in range, $ Q < Q_{\max}$, $\Lambda \geq \Lambda_{\min}$
L3 screen	10^3	0.1–1 s	Geometry converged, $F_{\text{rms}} < F_{\text{tol}}$
L4 screen	10^2	1–10 s	Bead mapping residual $< \delta$, phase coherent
L5 report	10^1	10–100 s	CAD geometry, corrosion model, integrity check

This is the Hourglass model applied at the platform level (§R). Each layer is a gate. The funnel is by design.

§2 L1 — Paper Identity

L1 — Paper Identity

Formula, class, name. Static. Never mutates.

A candidate that fails L1 is never instantiated at L2 or above.

2.1 What L1 carries

- **Formula** — Hill-order chemical formula string ("Fe2O3", "UO2", "Ti6Al4V")
- **IUPAC name** — systematic name (may be empty for elements)
- **Common name** — engineering or mineral name ("hematite", "urania", "sphalerite")
- **CAS number** — registry identifier
- **Class tag** — coarse classification: "element", "oxide", "halide", "organometallic", "alloy", "mineral"
- **Z-set and stoichiometry** — atomic numbers and coefficients

2.2 L1 gate

A species passes L1 if and only if:

1. Formula is non-empty and parseable
2. All $Z_i \in [1, 118]$
3. `class_tag` is not in the exclusion list

L1 costs microseconds. It is the first and cheapest filter. Run it on the full candidate set before doing anything else.

2.3 Process simulation context

In a full fuel cycle workflow, L1 represents the catalog:

- Every ore mineral has an L1 record ("Fe2O3", "FeS2", "FeCO3")
- Every reagent has an L1 record ("H2SO4", "NaOH", "HNO3")
- Every product target has an L1 record ("Fe", "UO2", "ThO2")

The L1 catalog is populated once and queried throughout the simulation. No physics is required at this stage.

§3 L2 — Atomistic Layer

L2 — Atomistic

Z, A, Q, ε . Physics and energy state.

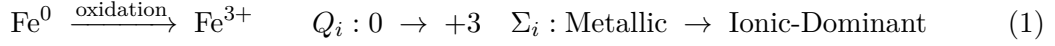
Environment-dependent. The §0 Identity Vector lives here.

3.1 What L2 carries

Field	Symbol	Units	Notes
Atomic number	Z_i	—	Immutable. Anchor of propagation invariant.
Mass number	A_i	—	Immutable. 0 = natural-abundance average.
Charge	Q_i	e	Environment-dependent. Fe^{3+} vs Fe^0 .
LJ well depth	ε_i	kcal/mol	Energy state. Scales with polarisability.
LJ radius	σ_i	Å	Size parameter.
Temperature	T_i	K	Local thermal state.
Pressure	P_i	bar	Local mechanical state.
Phase	ϕ_i	—	SOLID, LIQUID, GAS, AQUEOUS, ...
Σ_i	StructuralRole	—	Bonding topology prior (§0).
Λ_i	StabilityClass	—	Persistence class (§0).

3.2 The environment-dependence of Q and ε

This is not an approximation — it is physics. Fe^{3+} in sulphuric acid leach solution behaves fundamentally differently from Fe^0 in a solid ore body:



The ε track this transition:

- Fe^0 : $\varepsilon \approx 0.13$ kcal/mol (metallic, dispersion-dominant)
- Fe^{3+} (aq): $\varepsilon \approx 0.55$ kcal/mol (ionic, Coulomb-dominant)

Your ε field tracks that energy transition at every simulation step.

3.3 L2 gate

Candidates fail L2 if:

- $\varepsilon < 0$ or $\varepsilon > \varepsilon_{\max}$ (unphysical energy state)
- $|Q_i| > Q_{\max}$ (exceeds ionic saturation)
- $\Lambda_i < \Lambda_{\min}$ (unstable in this environment)

§4 L3 — Molecular 3D Layer

L3 — Molecular 3D

Geometry, bonding, coordination, local environment.
Built by FIRE minimisation on the L2 state.

4.1 What L3 carries

- Optimised 3D positions (Å) for all atoms in the species
- Bond table: pairs (i, j) , order, length, bond energy proxy
- Angle table: triples (i, j, k) with deviation from ideal
- Coordination record per atom: coordination number, geometry class, bond-length spread (distortion indicator)
- Quality metrics: F_{rms} , total energy, convergence flag
- Local environment summary: mean coordination, anisotropy index

4.2 L3 in the process simulation

The acid leach stage is an L3 event. When Fe_2O_3 dissolves in H_2SO_4 :

1. L1: "Fe2O3" + "H2SO4" are both in the catalog
2. L2: $Q(\text{Fe})$ increments from $0 \rightarrow +3$ over the dissolution timestep
3. L3: Fe–O bonds break; Fe–O(H_2O) coordination shell forms
4. L4: The solvated Fe^{3+} complex becomes a charged CG bead

L3 is where the bonding topology changes are made explicit.

§5 L4 — Atomistic Bead Layer

L4 — Atomistic Bead (CG)

Coarse-grained beads, phase behaviour, bulk interactions.
Anisotropic surface mapping lives here.

5.1 What L4 carries

- The full **Bead** struct from the CG layer, including:
 - Σ_i (structural role) and Λ_i (stability class)
 - Three-channel role-weighted interaction kernel weights
 - Anisotropic surface descriptor (Fuzzy Ball Model)
 - Slow environmental state η from the 6+9 Stepper
- Phase region: dominant phase + volume fraction + order parameter P_2
- Local density ρ_B and coordination number C_B
- Back-reference to the dominant L2 atom (preserving Z, A, Q)

5.2 Anisotropic surface mapping

The anisotropic surface descriptor is the Fuzzy Ball Model (§Fuzzy). At L4, the bead's surface carries a three-channel spherical harmonic field:

$$S^{(k)}(\theta, \phi) = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{+\ell} c_{\ell m}^{(k)} Y_{\ell}^m(\theta, \phi), \quad k \in \{\text{steric, electrostatic, dispersion}\}$$

This is the Container Geometry Approximation Ω_i from §0, made concrete.

§6 L5 — Macro / CAD Layer

L5 — Macro / CAD

Actual geometry, volumes, surfaces, material bodies.
A pipe. A reactor wall. A valve seat.

6.1 What L5 carries

Field	Units	Description
<code>body_type</code>	—	Pipe, Vessel, Valve, HeatExchanger, ...
<code>material_formula</code>	—	L1 formula of bulk material
<code>length_m</code>	m	Body length
<code>outer_diameter_m</code>	m	Outer diameter
<code>wall_thickness_m</code>	m	Wall thickness
<code>volume_m3</code>	m ³	Total enclosed volume
<code>surface_area_m2</code>	m ²	Wetted or exposed surface area
<code>corrosion_depth_m</code>	m	Wall loss from chemical attack
<code>oxide_layer_m</code>	m	Passivation / scale layer thickness
<code>roughness_Ra_m</code>	m	Surface roughness
<code>dominant_species</code>	—	L1 formula of the surface species
<code>temperature_K</code>	K	Operating temperature
<code>pressure_bar</code>	bar	Operating pressure
<code>fluid_formula</code>	—	L1 formula of the process fluid
<code>structural_integrity</code>	0–1	1 = pristine, 0 = failed

6.2 Particle identity at L5

The iron in a pipe wall at L5 is still Fe ($Z = 26$) underneath. The macro geometry — wall thickness, corrosion depth, surface roughness — is the accumulated consequence of:

- L2: Fe^0 ε determining surface adhesion energy
- L3: Fe–O bond formation rate at the surface
- L4: oxide scale growth modelled by bead density and phase fields
- L5: wall loss recorded as `corrosion_depth_m`

The structural integrity field is a direct function of these upward-propagated identities:

$$\text{integrity} = 1 - \frac{\delta_{\text{corr}}}{t_{\text{wall}}} \in [0, 1]$$

where δ_{corr} is the corrosion depth and t_{wall} is the original wall thickness.

§7 The L2 L4 Hard Boundary

Hard Problem 1: The L2 L4 Phase Transition Boundary

This is where most simulations cheat or break.

Atomic behaviour (discrete particle physics, quantum-derived ε , explicit bonding) transitions into coarse-grained bulk behaviour (phase fields, density order parameters, collective interactions). It is not a sharp line. It is a controlled approximation zone.

7.1 What cheating looks like

1. **Silent ε switch.** Swapping from LJ ε (L2) to CG bead ε (L4) without recording the mapping error. The energy changes by an unmeasured amount.
2. **Charge loss.** Lumping Fe^{3+} and O^{2-} into a neutral bead loses the Coulomb interaction entirely. The electrostatic energy disappears from the simulation without a trace.
3. **Single-phase fiction.** Treating a bead group as single-phase when atoms in it straddle a liquid-solid boundary. Phase boundaries do not coincide with CG mapping boundaries.
4. **Frozen structural role.** Using the same $\Sigma_i = \text{Metallic}$ for Fe across the liquid-metal/ionic transition. The bonding topology changes; the role must follow.

7.2 The `layer_boundary.hpp` solution

Five boundary checks run at every L2→L4 transition:

Unit	Name	What it measures
B1	Energy consistency	$ E_{\text{L4}} - E_{\text{L2}} $ per bead group. Flags when CG energy deviates from atomistic.
B2	Charge conservation	$ q_{\text{L4}} - \sum_j q_j^{\text{L2}} $ per bead. Non-conservation above threshold is always flagged.
B3	Phase coherence	Detects bead groups spanning multiple phases. Optional split or flag.
B4	Structural role re-evaluation	Re-classifies Σ_i from current Q_i and Z_i . Updates the bead in-place if changed.
B5	Mapping residual	Weighted composite ($0.5 \times E_{\text{term}} + 0.3 \times Q_{\text{term}} + 0.2 \times \phi_{\text{term}}$). High residual > 0.15 is flagged.

The boundary report is a first-class output. Every approximation is recorded with its magnitude. Nothing is silent.

7.3 The B4 role re-evaluation equation

$$\Sigma_i^{(\text{boundary})} = \begin{cases} \text{IonicDominant} & Z_i \in \text{transition metals} \wedge |Q_i| > 0.8 \\ \text{Metallic} & Z_i \in \text{transition metals} \wedge |Q_i| \leq 0.8 \\ \text{IonicDominant} & |Q_i| > 0.5 \\ \text{DirectionalCovalent} & \text{otherwise} \end{cases} \quad (2)$$

This is equation (??) made computational.

§8 Hard Problem 2: Compute Cost

Hard Problem 2: Compute Cost

Pushing millions of candidates through all five layers is not feasible without aggressive early rejection and lazy evaluation.

8.1 The answer: lazy layer evaluation

No particle is evaluated at layer k until it has passed the gate at layer $k - 1$. The `LayerParticle` struct uses `std::optional` for L3, L4, and L5 precisely for this reason:

```
struct LayerParticle {
    L1_PaperIdentity          11;    // always populated
    L2_AtomicState            12;    // populated after L1 gate
    std::optional<L3_MolecularGeometry> 13;    // populated after L2 gate
    std::optional<L4_AtomicBeadState> 14;    // populated after L3 gate
    std::optional<L5_MacroGeometry>    15;    // populated after L4 gate
};
```

L3, L4, and L5 are never heap-allocated for particles that fail their predecessor gate. The rejected candidate costs only its screen cost, not the full simulation cost.

8.2 The answer: Hourglass at platform scale

The Hourglass Model (§R) was designed for individual configuration sampling. At platform scale the same logic applies to the candidate population:

1. **Mouth:** Generate 10^6 L1 candidates from combinatorial formula space
2. **Neck:** Sequential L1/L2/L3 gates reduce to $\sim 10^2$
3. **Base:** Surviving candidates get full L4/L5 treatment and are ranked by composite score

The gate criteria are the same physics as the layer boundaries. They are not ad hoc filters — they are the physics screening.

§9 Full Fuel Cycle Simulation

9.1 Discovery end

Generate candidate metal species, classify, screen. Millions of candidates, most rejected at L1, survivors get full L2–L5 treatment. This document is the routing specification for that pipeline.

Stage	Input	Output
Candidate generation	L1 formula space	10^6 L1 records
L1 screen	Class + formula validity	$\sim 10^4$ survivors
L2 evaluation	ε , Q , Λ screening	$\sim 10^3$ survivors
L3 geometry	FIRE minimisation	$\sim 10^2$ converged structures
L4 mapping	CG bead + phase + anisotropy	$\sim 10^1$ viable materials
L5 report	CAD geometry, corrosion, integrity	Product candidates

9.2 Process simulation

Ore input $\rightarrow$ crushing $\rightarrow$ dissolution $\rightarrow$ separation $\rightarrow$ refining. Every stage is a chemical environment. Species identities evolve as they move through:

Stage	Active layer(s)	Key identity change
Ore body	L1, L4	Fe as Fe ₂ O ₃ , FeS ₂
Crushing / grinding	L5	Surface area increases
Acid leach	L2, L3	Fe ³⁺ forms; Q , Σ update
Solvent extraction	L3, L4	Complex speciation; bead re-mapping
Stripping	L2, L3	Metal ion concentration
Electrowinning	L2, L5	Metal deposits; cathode grows
Refining	L3, L4, L5	Alloy composition; grain structure

9.3 Pipe and reactor simulation

L3 to L5. Material integrity, corrosion, fluid-wall interactions, thermal loads. The CAD layer is the physical infrastructure. The particle layer is what is eating it or flowing through it.

The corrosion loop:

1. L2: $\varepsilon(\text{Fe})$ and $Q(\text{Fe})$ at the pipe surface
2. L3: Fe–O bond formation rate (surface oxidation kinetics)
3. L4: Oxide scale growth rate (bead phase field, η slow state)
4. L5: `corrosion_depth_m` increments; `structural_integrity` decrements

Conclusion

The five-layer stack is the architectural spine of the VSEPR-SIM materials discovery platform. Its properties:

1. **Vertically integrated.** The same particle identity (Z , A , Q) propagates from ore body to finished material without discontinuity.
2. **Lazy.** Candidates are never evaluated at cost above their gate status. The funnel is the architecture.
3. **Explicit at every boundary.** The L2L4 hard boundary is not approximated silently. `layer_boundary.hpp` records every energy gap, every charge error, every role update, every phase mismatch.
4. **Anti-black-box.** Every layer transition is a named, typed, inspectable C++ struct. Every approximation has a magnitude. Nothing is hidden.
5. **Deterministic.** Same candidate + same environment + same parameters $\Rightarrow$ same layer traversal $\Rightarrow$ same output.